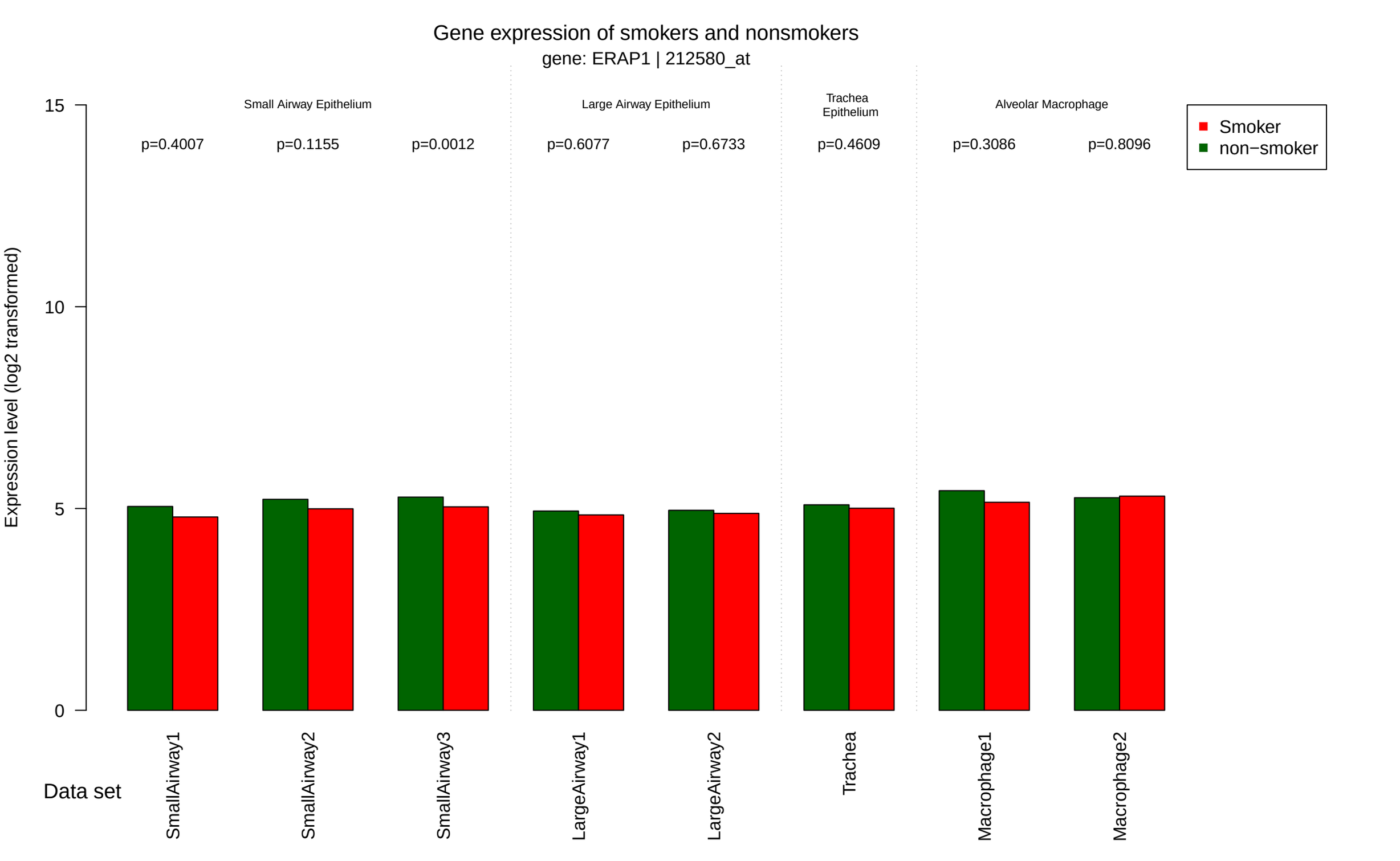
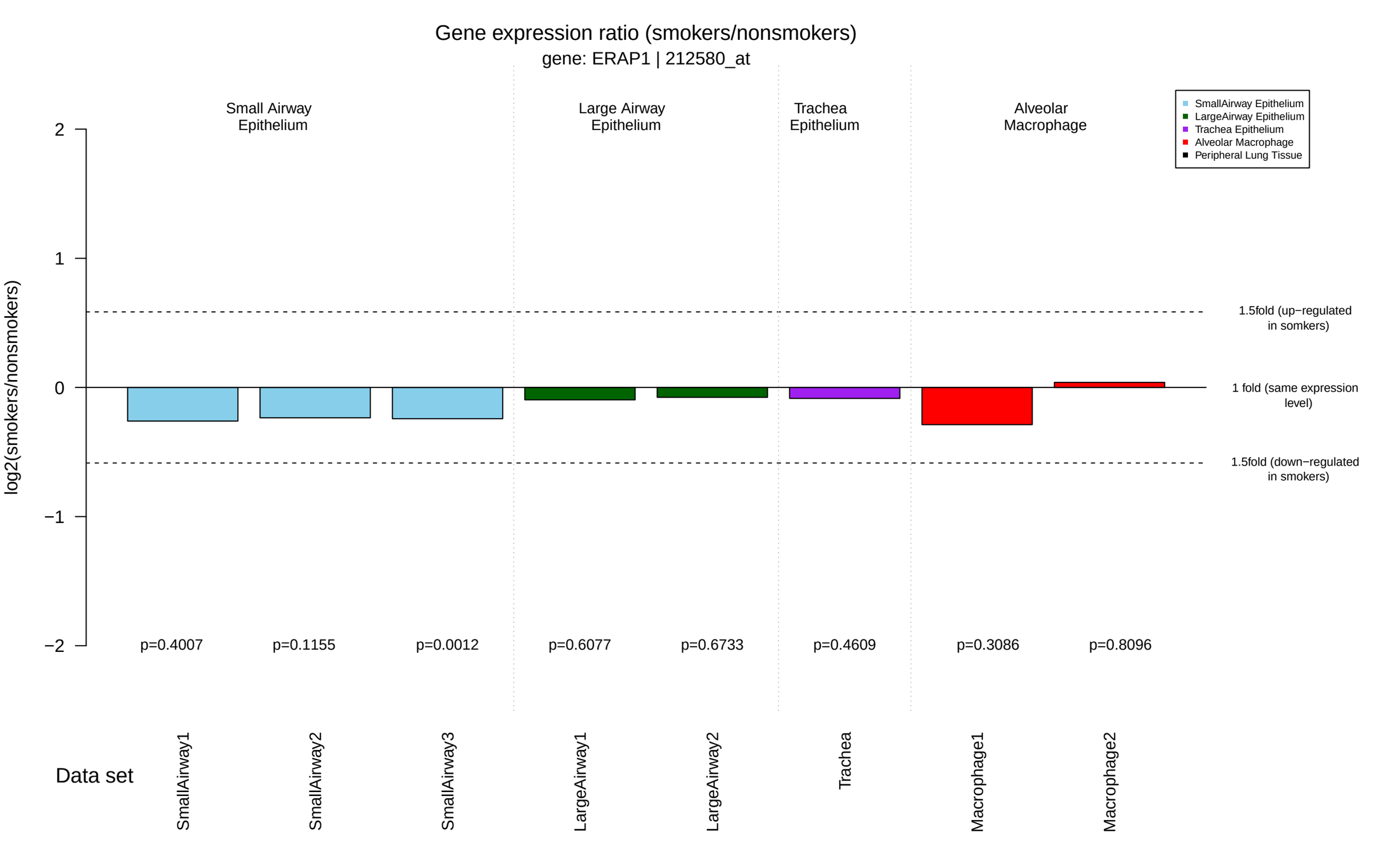
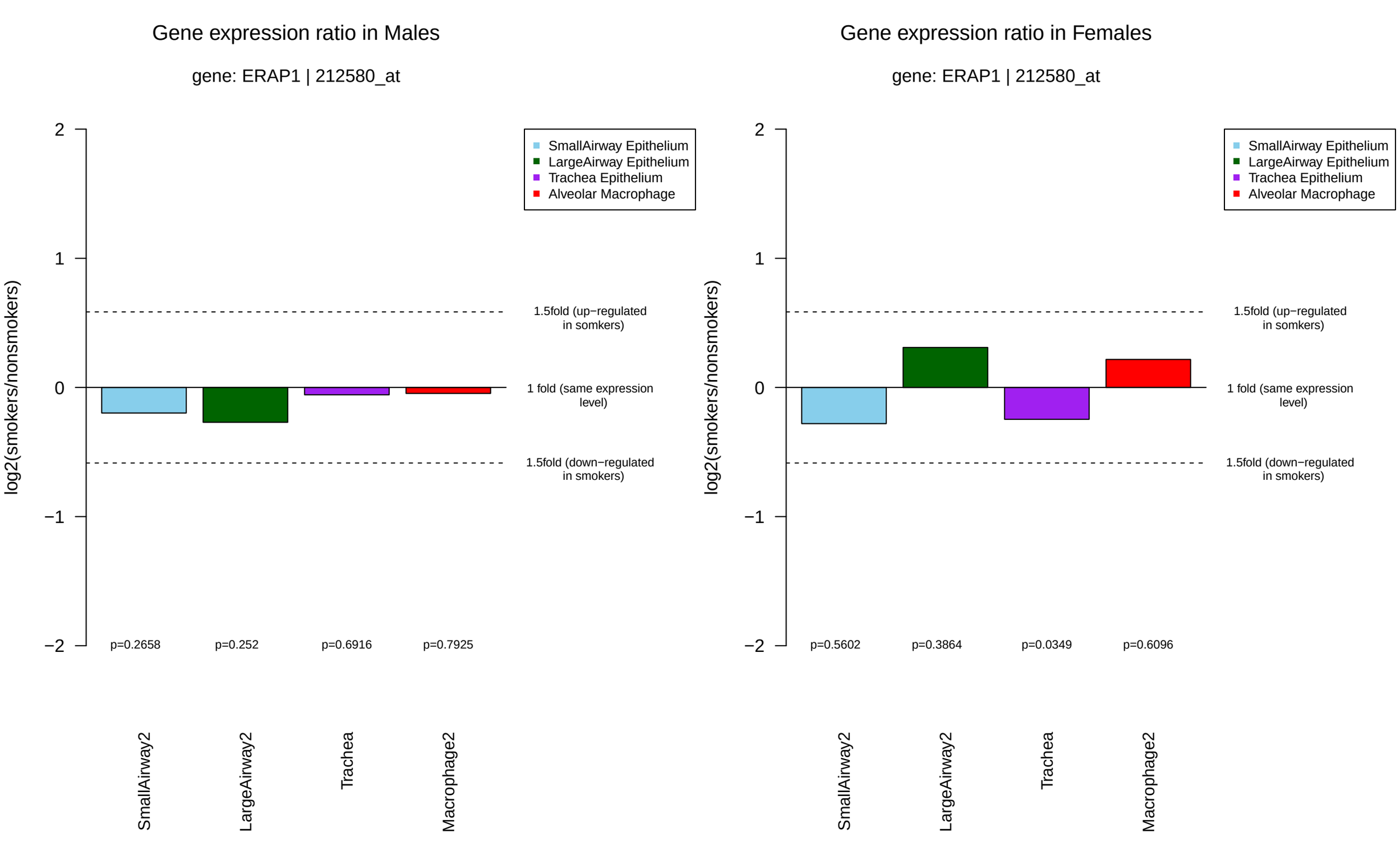
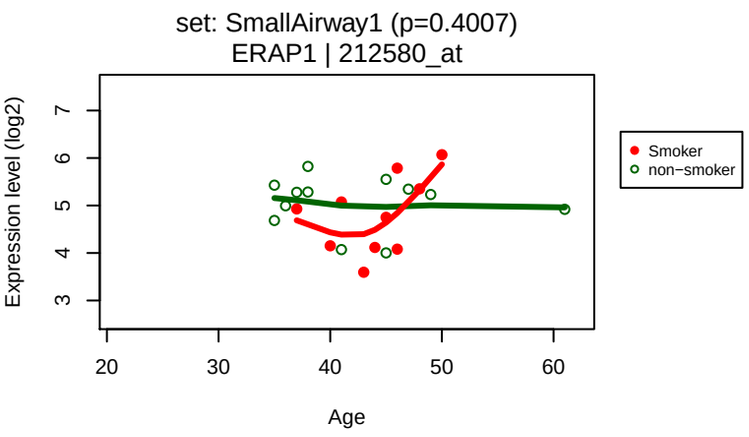
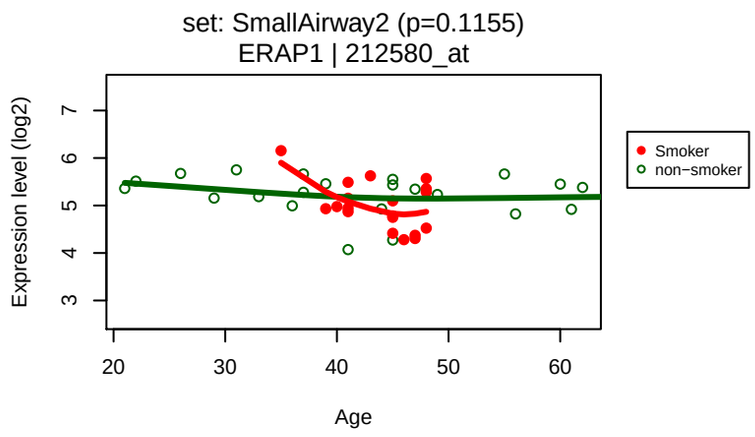
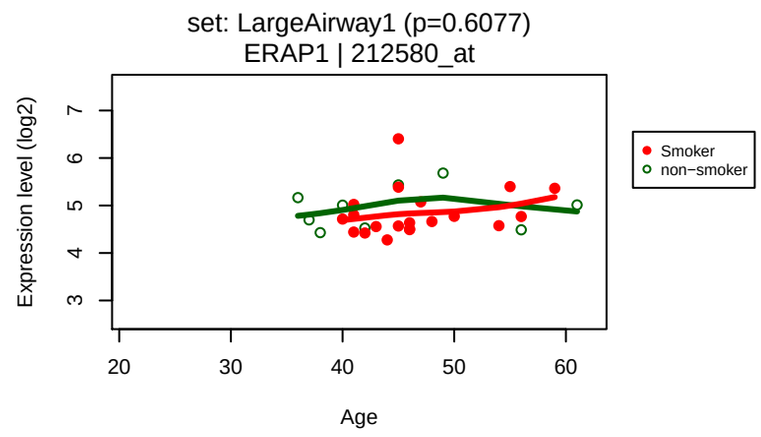

Gene expression of smokers and nonsmokers
gene: ERAP1 | 212580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4007
p=0.1155
p=0.0012
p=0.6077
p=0.6733
p=0.4609
p=0.3086
p=0.8096
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

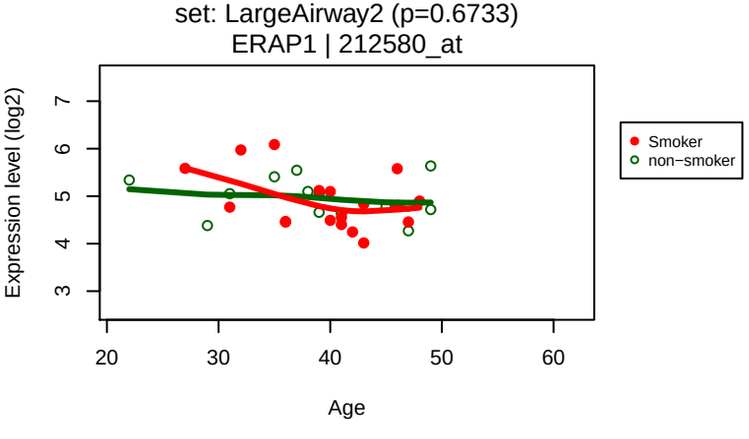

Gene expression ratio (smokers/nonsmokers)
gene: ERAP1 | 212580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4007
p=0.1155
p=0.0012
p=0.6077
p=0.6733
p=0.4609
p=0.3086
p=0.8096
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

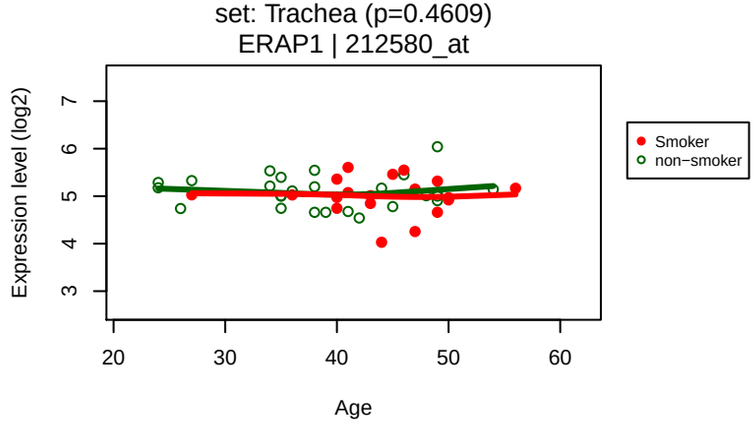

Gene expression ratio in Males
gene: ERAP1 | 212580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2658
p=0.252
p=0.6916
p=0.7925
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERAP1 | 212580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5602
p=0.3864
p=0.0349
p=0.6096
SmallAirway2
LargeAirway2
Trachea
Macrophage2

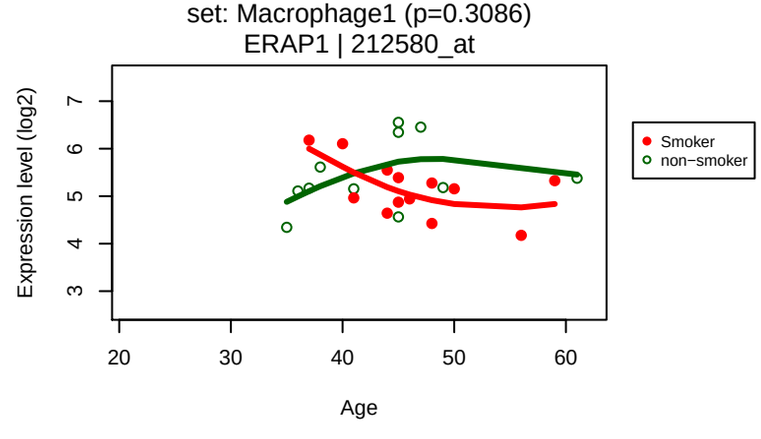

set: SmallAirway1 (p=0.4007)
ERAP1 | 212580_at
Expression level (log2)
Age
Smoker
non−smoker

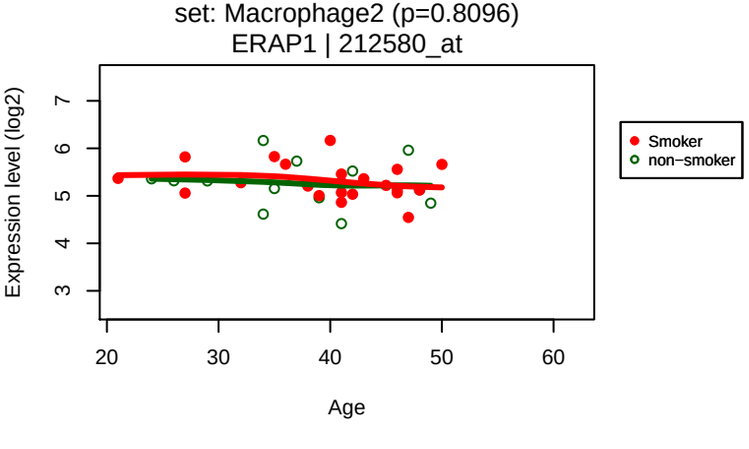

set: SmallAirway2 (p=0.1155)
ERAP1 | 212580_at
Expression level (log2)
Age
Smoker
non−smoker

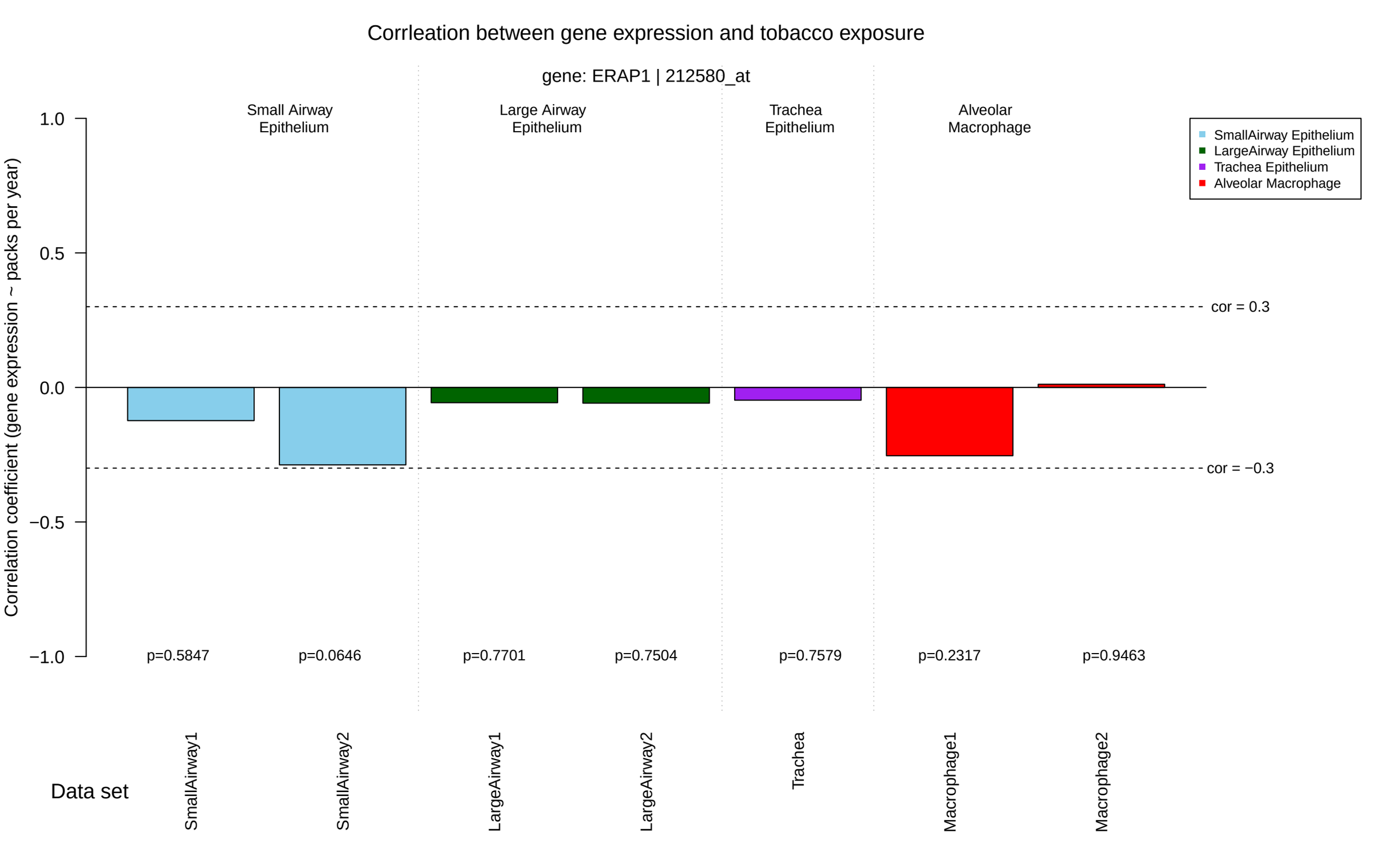

set: LargeAirway1 (p=0.6077)
ERAP1 | 212580_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6733)
ERAP1 | 212580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4609)
ERAP1 | 212580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3086)
ERAP1 | 212580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8096)
ERAP1 | 212580_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERAP1 | 212580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5847
p=0.0646
p=0.7701
p=0.7504
p=0.7579
p=0.2317
p=0.9463
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2